



**EVALUATION OF TEST ITEMS EFFICACY IN PROMOTING STROKE
REHABILITATION IN THE RAT STROKE MODEL (T-MCAO)**

Outline Protocol

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1. ABBREVIATIONS:

AAALAC	Association for Assessment and Accreditation of Laboratory Animal Care
BSW	Body Swing test
CCA	Common Carotid Artery
ECA	External Carotid Artery
FLP	Forelimb Placing test
HEPA	High Efficiency Particulate Air
ICA	Internal Carotid Artery
NSS	Neurological Score Scale
OECD	Organization for Economic Co-operation and Development
SC	Subcutaneous
SD	Sprague Dawley
SOP	Standard Operating Procedure
ST	Stepping Test
t-MCAO	transient Middle Cerebral Artery Occlusion

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2. **STUDY COMPLIANCE**

These studies are performed at Pharmaseed based on the OECD principles of Good Laboratory Practice ENV/MC/CHEM (98)17, and following Pharmaseed's SOPs.

3. **OBJECTIVES**

The objective of this outline is to present the details of the procedures for evaluating Test Items' efficacy in promoting stroke rehabilitation, in the rat transient Middle Cerebral Carotid Occlusion (t-MCAO) model.

3.1 Study end points

- mNSS score (Neuroscore)
- Stepping Test (ST)
- Body Swing Test (BSW)
- Forelimb Placing Test (FLP)
- Body weight measurements
- Brains harvesting

4. **EXPERIMENTAL MODEL**

4.1 Animals

4.1.1 Species/Strain

Rats/SD

4.1.2 Gender / Age

Males / ~3 months at study initiation

4.1.3 Number

Depending on the number of Study Groups (see below).

4.1.4 Body weight

Approximately 300 g at study initiation. The minimum and maximum weight within the group should be within a range of $\pm 20\%$ of the group's mean weight.

4.1.5 Acclimation period:

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Not less than five days.

4.1.6 Identification

Three position ear notching and cage cards.

4.2 Animal Management

4.2.1 Housing

Animal handling is performed according to guidelines of the National Institute of Health (NIH) and the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC).

4.2.2 Diet

Animals are fed *ad libitum* a commercial rodent diet and have free access to acidified drinking water.

4.2.3 Contaminants

Reasonably expected contaminants in food and water supplies should not have the potential to influence the outcomes of this test.

4.2.4 Environment conditions

Animals are housed under standard laboratory conditions, air-conditioned and filtered (HEPA F6/6) with adequate fresh air supply (minimum 15 air changes/hour). Animals are kept in a climate-controlled environment. Temperatures range is 20–24°C and relative humidity range is 30–70% with a 12 hours light and 12 hours dark cycle.

4.2.5 Allocation

Animals are allocated into the test groups, according to the Neuroscore results on Day 2, in order to have similar distribution of rat's performance between the groups.

4.3 Veterinary care

Animals are inspected upon arrival in order to determine being fit for the study. Animals are inspected daily for any signs of morbidity or mortality.

4.4 Ethics Committee

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This study should be performed after approval by the appropriate "Ethics Committee" and in compliance with "The Animal Welfare Act".

5. **EXPERIMENTAL DESIGN AND CONDITIONS**

5.1 **Animals Welfare**

5.1.1 Pain reducing methods

Analgesia: Buprenorphine (Opioids) or Carprofen (NSAID).

Buprenorphine

Animal	Dose (mg/kg)	Route	Duration
Rat	0.05-0.1	SC	6-12 hr

Carprofen

Animal	Dose (mg/kg)	Route	Duration
Rat	5	SC	12-24 hr

5.2 **Humane Endpoints**

Animals found in a moribund condition, animals showing severe pain and enduring signs of severe distress (such as dyspnea, lateral recumbence, convulsions, plegia or inability to reach food or water), and animals showing a decrease of body weight larger than 20% from initial body weight are humanely euthanized.

5.3 **Study initiation definition**

t-MCAO procedure Day is defined as "DAY 1" of the study. See [Table 2](#) for the detailed study timeline.

5.4 **PROCEDURES**

5.4.1 **Anesthesia**

On the day of surgery anesthesia should be induced with 4% isoflurane in a mixture of 70% N₂O and 30% O₂ and maintained with 1.5-2% isoflurane, and the animals are kept on a heating pad during the surgery. Analgesic treatment is given before surgery, on the end of Day 1 and during the next 4 days, while watching the recovery of the wound.

5.4.2 **Surgical procedure**

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Transient middle cerebral artery occlusion is performed according to the method described R. Schmid-Elsaesser et al. (see [Ref. 1](#)). The right Common Carotid Artery (CCA) is exposed through a midline neck incision and carefully dissected free from surrounding nerves and fascia – from its bifurcation to the base of the skull. The occipital artery branches of the External Carotid Artery (ECA) are then isolated, and these branches are dissected and coagulated. The ECA is dissected further distally and coagulated along with the terminal lingual and maxillary artery branches, which is then divided. The Internal Carotid Artery (ICA) is isolated and carefully separated from the adjacent vagus nerve, and the pterygopalatine artery is ligated close to its origin with a 5-0 nylon suture (SMI, Belgium). Next a 4-0 silk suture is tied loosely around the mobilized ECA stump, and a 4 cm length of 4-0 monofilament nylon suture (the tip of the suture is blunted by using a flame, and the suture is coated with polylysine, prior to insertion) is inserted through the proximal ECA into the ICA and thence into the circle of Willis, effectively occluding the MCA. The surgical wound is closed and the animals are returned to their cages to recover from anesthesia. One and a half hours after occlusion rats are re-anesthetized, monofilament is withdrawn to allow reperfusion, surgical wound is closed and the rats are returned to their cages.

5.4.3 Neuroscore and Exclusion Criteria

Animals are subjected to the mNSS test (Neuroscore) on Day 2, 24 hours post reperfusion. Only animals with an overall score of ≥ 10 are included in this study ([See Appendix](#)).

5.5 Test Item administration

Test Item administration and schedule depends on the investigated drug

See [Table 1](#) for Group Allocation.

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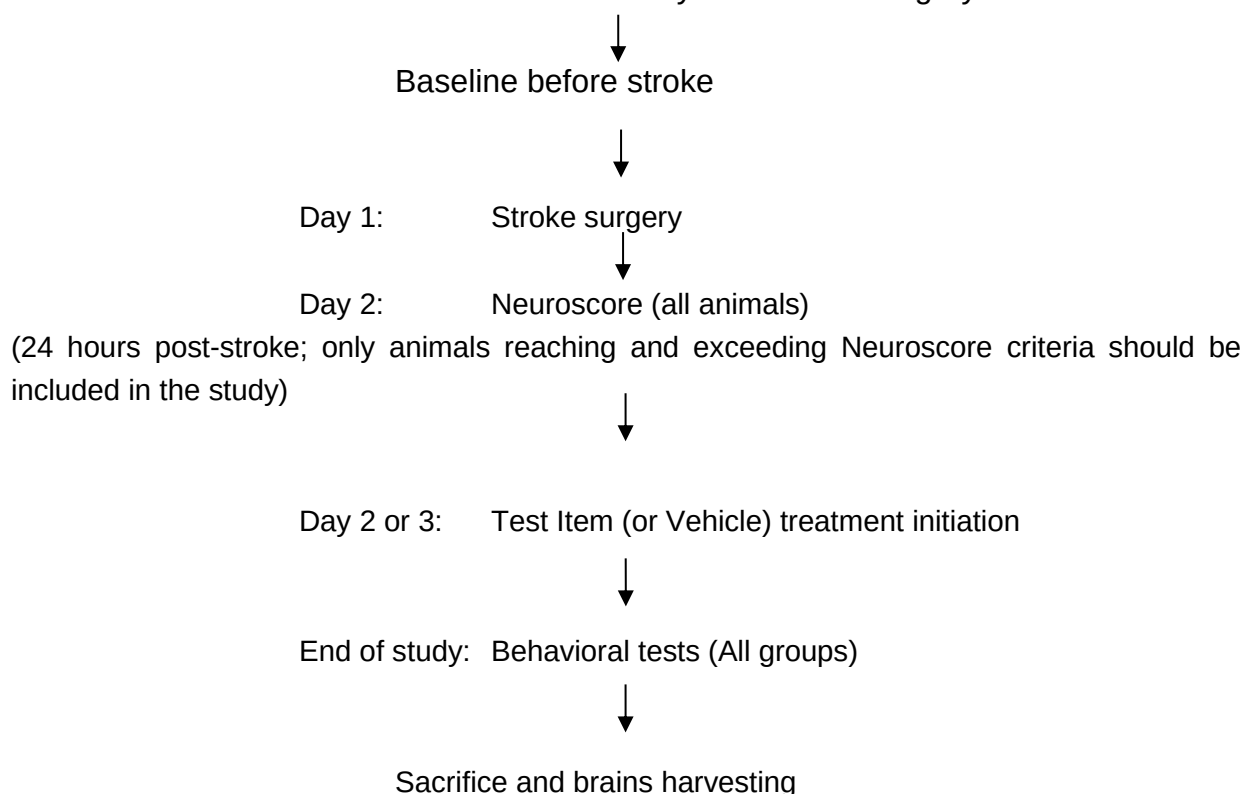
Table 1: An example for groups allocation

Group*	Treatment	Dose	Administration	Treatment duration (days)	Initial # of rats	Total # of rats (final)
1	Vehicle	0	Depending on the drug	Depending on the drug	25	15
2	Test Item	Low dose			25	15
3	Test Item	Mid dose			25	15
4	Test Item	High dose			25	15

**The study should be performed in consecutive and balanced cycles, resulting in at least 15 rats per study group. Vehicle treated rats should be allocated to each of the cycles.*

5.6 Handling, surgery, and behavioral tests timetable:

Handle the animal's minimum for five days before the surgery



5.7 Body weight

Body weight is measured three times a week.

5.8 Mortality

Observations for signs of morbidity and mortality are performed twice daily. Moribund animals or animals obviously in pain or showing signs of severe and enduring distress are euthanized. The time of death should be recorded as precisely as possible.

5.9 Post-Operative Monitoring

Following surgery, animals remain on the heating pad until they are awake from the anesthesia. They are observed frequently on the day of MCAO surgery (Day 1) and at least once a day thereafter.

- **Behavioral test details**

Behavioral evaluations are performed by investigators blinded to treatment assignment.

5.10 PRE-DOSING BEHAVIOUR HABITUATION TESTING:

5.10.1 Limb Placing Evaluation:

Before operation and on End of Study

The limb placing tests are divided into forelimb and hindlimb tests. For the forelimb-placing test, the examiner holds the rat close to a tabletop and scores the rat's ability to place the forelimb on the tabletop in response to whisker, visual, tactile, or proprioceptive stimulation. Similarly, for the hindlimb placing test, the examiner assesses the rat's ability to place the hindlimb on the tabletop in response to tactile and proprioceptive stimulation. Separate sub-scores are obtained for each mode of sensory input and added to give total scores (for the forelimb placing test: 0=normal, 12=maximally impaired; for the hindlimb placing test: 0=normal; 6=maximally impaired). Scores are given in half-point increments (see below). Typically, there is a slow and steady recovery of limb placing behavior during the first month after stroke.

Forelimb placing test (0-12):

Whisker placing (0-2);

Visual placing (forward (0-2), sideways (0-2))

Tactile placing (dorsal (0-2), lateral (0-2))

Proprioceptive placing (0-2).

5.10.2 Stepping Test Evaluation:

Before operation and on End of Study

Animals are tested for forelimb akinesia in the Stepping Test (ST), according to the following procedure: the animal is held with its hind limbs fixed in one hand and the forelimb, not to be monitored, in the other, while the unrestrained fore-paw touches the table. The number of adjusting steps are counted while the animal is moved sideways along the table surface (85 cm during approximately five seconds), in the forehand & backhand direction for both forelimbs.

5.10.3 Body swing test Evaluation:

Before operation and on End of Study

The rat is held approximately one inch from the base of its tail. It is then elevated to an inch above a surface of the table. The rat is held in the vertical axis, defined as no more than 10° to either the left or the right side. A swing is recorded whenever the rat moves its head out of the vertical axis to either side. Before attempting another swing, the rat must return to the vertical position for the next swing to be counted. Twenty (20) total swings are counted. A normal rat typically has an equal number of swings to either side. Following focal ischemia, the rat tends to swing to the contralateral side (left side in this case). Body swing scores are expressed as a percentage of rightward over total swings. There is a spontaneous partial recovery of body swing scores (toward 50%) during the first month after stroke induction.

5.10.4 Neurological scoring (NSS):

The Modified Neurological Rating Scale (mNRS), or Neuroscore, is performed before operation, on Day 2 and on End of Study

The individual performing the behavioral assessments is unaware of the drug/dose given (blinded test). Neuroscore performed according to the NSS table (total score of 18 – see [Appendix](#)) ([Ref. 2](#)).

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5.11 Sacrifice and Samples Collection:

All animals – On the End of the Study, rats are sacrificed by CO₂ inhalation. Brains should be harvested following routine perfusion.

Table 2: Study Plan

Study Time Line Procedure	Before surgery	1	2	Depending on the investigated drug	End of Study	
Surgery		√				
Neuroscore	√		√		√	
Treatment				√		
Body swing test	√				√	
Limb placing	√				√	
Stepping test	√				√	
Termination						√
Brain havasting						√

6. RESULTS

6. 1 Data compilation

Data should be summarized in a tabular form showing for each group the number of animals used, the individual results of body weights, tissues weight, the different behavioral scores and any other relevant information. Summary figures and tables should also be included.

6.2 Statistical analysis

Numerical results are given as means ± standard errors. If applicable, statistical analysis is carried out using one-way or two-way ANOVA. A probability of 5% ($p \leq 0.05$) should be regarded as significant.

7. **REFERENCES**

- 7.1 Schmid-Elsaesser R., Zausinger S., Hungerhuber E., Baethmann A., Reulen H.J. A critical reevaluation of the intraluminal thread model of focal cerebral ischemia: evidence of inadvertent premature reperfusion and subarachnoid hemorrhage in rats by laser-Doppler flowmetry. *Stroke*. 1998 ;29(10):2162-2170.
- 7.2 Modo M., Stroemer R.P., Tang E., Veizovic T. Sowniski P., Hodges H. Neurological sequelae and long-term behavioural assessment of rats with transient middle cerebral artery occlusion. *Journal of Neuroscience Methods*. 2000: 194:99-109.

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8. APPENDIX

Neuroscore: mNSS – modified Neurological Scoring Scale (Total score: 18)

Neurological Severity Score											
Study No.:								Date:			
Group No.:											
Animal No.:											
Motor Test											
Flexion of forelimb	0/1										
Flexion of hind limb	0/1										
Head moved >10° to vertical axis within 30 s	0/1										
Normal walk	0										
Inability to walk straight	1										
Circling toward paretic side	2										
Falls down to paretic side	3										
Sensory Test											
Placing test (visual and tactile test)	0/1										
Proprioceptive test (deep sensation, pushing paw against table edge to stimulate limb muscles)	0/1										
Beam Balance Test											
Balances with steady posture	0										
Grasps side of beam	1										
Hugs beam and 1 limb falls down from beam 2	2										
Hugs beam and 2 limbs fall down from beam, or spins on beam (60 s)	3										
Attempts to balance on beam but falls off (>40 s)4	4										
Attempts to balance on beam but falls off (>20 s)5	5										
Falls off; no attempt to balance or hang on to beam (<20 s)	6										
Reflex Absence and Abnormal Movements											
Pinna reflex (head shake when auditory meatus is touched)	0/1										
Corneal reflex (eye blink when cornea is lightly touched with cotton)	0/1										
Startle reflex (motor response to a brief noise from snapping a clipboard paper)	0/1										
Seizures, myoclonus, myodystony	0/1										
Maximal Total Score	18										